

TABLE 58-1 Prediction of Compensatory Responses to Simple Acid-Base Disturbances and Pattern of Changes

DISORDER	PREDICTION OF COMPENSATION	RANGE OF VALUES		
		PH	HCO ₃ ⁻	Paco ₂
Metabolic acidosis	$\text{Paco}_2 = (1.5 \times \text{HCO}_3^-) + 8 \pm 2$ <i>or</i> Paco ₂ will ↓ 1.25 mmHg per mmol/L ↓ in [HCO ₃ ⁻] <i>or</i> $\text{Paco}_2 = [\text{HCO}_3^-] + 15$	Low	Low	Low
Metabolic alkalosis	Paco ₂ will ↑ 0.75 mmHg per mmol/L ↑ in [HCO ₃ ⁻] <i>or</i> Paco ₂ will ↑ 6 mmHg per 10 mmol/L ↑ in [HCO ₃ ⁻] <i>or</i> $\text{Paco}_2 = [\text{HCO}_3^-] + 15$	High	High	High
Respiratory alkalosis		High	Low	Low
Acute	[HCO ₃ ⁻] will ↓ 0.2 mmol/L per mmHg ↓ in Paco ₂			
Chronic	[HCO ₃ ⁻] will ↓ 0.4 mmol/L per mmHg ↓ in Paco ₂			
Respiratory acidosis		Low	High	High
Acute	[HCO ₃ ⁻] will ↑ 0.1 mmol/L per mmHg ↑ in Paco ₂			
Chronic	[HCO ₃ ⁻] will ↑ 0.4 mmol/L per mmHg ↑ in Paco ₂			

1. Range of values header

WHAT YOU SEE

RANGE OF VALUES columns: pH | HCO3 | PaCO2

WHAT IT ENCODES

The systematic ABG approach: (1) check pH to name acidemia (<7.35) vs alkalemia (>7.45); (2) check whether PaCO2 (normal 40 ± 5) or HCO3 (normal 24 ± 2) explains the pH shift to identify the PRIMARY disorder; (3) use the compensation formula to predict the expected secondary value; (4) calculate the anion gap on every metabolic acidosis. The primary disturbance and its compensation always move pH in OPPOSITE directions from each other, but the primary and compensatory variables (HCO3 and PaCO2) move in the SAME direction. Overcompensation past normal pH never happens with a single disorder — if pH is normal but PaCO2 and HCO3 are both deranged, suspect a mixed disorder.

BOARD TRAP

WRONG answer: calling a derangement "fully compensated" when pH has crossed to the other side of 7.40. Discriminator: compensation never normalizes or overshoots pH — if a patient with chronic respiratory acidosis is frankly alkalemic, a second primary metabolic alkalosis is present (e.g., overdiuresis in COPD).

2. Metabolic acidosis — Winter's formula

WHAT YOU SEE

Metabolic acidosis: PaCO2 = (1.5 × HCO3) + 8 ± 2

WHAT IT ENCODES

Winter's formula gives the expected respiratory compensation (the PaCO2 the lungs should drive to via hyperventilation) for any measured HCO3 in metabolic acidosis: expected PaCO2 = 1.5 × HCO3 + 8 ± 2. Example: HCO3 of 12 → expected PaCO2 = 1.5(12) + 8 = 26 ± 2 (24–28). Hyperventilation (Kussmaul respirations in DKA) blows off CO2 within minutes to hours and is maximal by ~12–24h, but PaCO2 rarely falls below ~10 mmHg. Always pair Winter's with an anion gap and a delta-delta to fully characterize the disorder.

BOARD TRAP

WRONG answer: assuming a low PaCO2 in a metabolic acidosis is automatically "appropriate" without doing the math. Discriminator: if measured PaCO2 is HIGHER than Winter's predicts → superimposed respiratory acidosis (e.g., DKA patient tiring/aspirating); if measured PaCO2 is LOWER than predicted → superimposed respiratory alkalosis (e.g., salicylate toxicity, sepsis, pregnancy).

3. Metabolic acidosis — alt rules

WHAT YOU SEE

PaCO2 will ↓ 1.25 mmHg per mmol/L ↓ in HCO3, or PaCO2 = HCO3 + 15

WHAT IT ENCODES

Bedside shortcuts equivalent to Winter's: (1) PaCO2 falls ~1.25 mmHg for each 1 mmol/L drop in HCO3 below 24; (2) PaCO2 ≈ HCO3 + 15; (3) the last two digits of the pH approximate the expected PaCO2 (e.g., pH 7.26 → PaCO2 ~26). These give a fast sanity check before reaching for the full Winter's calculation, but Winter's (± 2 range) remains the board-preferred answer.

BOARD TRAP

WRONG answer: using the "HCO3 + 15" rule outside its validity range. Discriminator: the rule only holds for HCO3 roughly 10–24; at very low HCO3 it diverges from Winter's, so when given a severe acidosis (HCO3 < 10) compute Winter's formula explicitly rather than trusting the shortcut.

4. Metabolic acidosis values (all Low)

WHAT YOU SEE

pH Low, HCO3 Low, PaCO2 Low

WHAT IT ENCODES

Primary fall in HCO3 lowers pH (acidemia); the lungs compensate by hyperventilating to lower PaCO2, partially restoring pH. HCO3 and PaCO2 move in the SAME direction (both low) — the signature of a metabolic acidosis with respiratory compensation. Next step is ALWAYS the anion gap (Na – [Cl + HCO3], normal 8–12): high-gap causes = MUDPILES/GOLDMARK (methanol, uremia, DKA, propylene glycol, INH/iron, lactate, ethylene glycol, salicylates), normal-gap (hyperchloremic) causes = diarrhea and renal tubular acidosis (urine anion gap separates them).

BOARD TRAP

WRONG answer: stopping at "metabolic acidosis with appropriate compensation" without the anion gap or delta-delta. Discriminator: both values low confirms direction only — a normal pH with low HCO3 and low PaCO2 plus a high gap can hide a triple disorder; compute the gap and delta-delta on every metabolic acidosis.

5. Metabolic alkalosis — formula

WHAT YOU SEE

Metabolic alkalosis: PaCO2 will ↑ 0.75 mmHg per mmol/L ↑ in HCO3, or ↑6 per 10 mmol/L ↑, or PaCO2 = HCO3 + 15

WHAT IT ENCODES

Expected respiratory compensation is hypoventilation that RAISES PaCO2: ~0.7 mmHg rise per 1 mmol/L rise in HCO3 (≈0.75 in this table, ≈6 per 10 mmol/L), giving expected PaCO2 ≈ 0.7 × HCO3 + 21, or the shortcut PaCO2 = HCO3 + 15. Example: HCO3 of 36 → expected PaCO2 ≈ 0.7(36) + 21 ≈ 46. Distinguish saline-RESPONSIVE causes (urine Cl < 20: vomiting/NG suction, diuretics, contraction alkalosis — treat with normal saline) from saline-RESISTANT (urine Cl > 20: hyperaldosteronism, Cushing, severe hypokalemia, Bartter/Gitelman).

BOARD TRAP

WRONG answer: predicting unlimited CO2 retention to fully correct the alkalemia. Discriminator: compensatory hypoventilation is LIMITED because rising PaCO2 worsens hypoxemia, which then drives ventilation — PaCO2 rarely exceeds ~55 mmHg; a PaCO2 above that implies a superimposed respiratory acidosis rather than normal compensation.

6. Metabolic alkalosis values (all High)

WHAT YOU SEE

pH High, HCO3 High, PaCO2 High

WHAT IT ENCODES

Primary rise in HCO3 raises pH (alkalemia); the lungs compensate by hypoventilating to retain CO2. HCO3 and PaCO2 move in the SAME direction (both high) — the signature of metabolic alkalosis with respiratory compensation. The maintenance of metabolic alkalosis usually requires volume depletion, chloride depletion, or hypokalemia, all of which sustain renal bicarbonate reabsorption; correcting these (saline + KCl) is therapeutic.

BOARD TRAP

WRONG answer: attributing a high PaCO2 with high HCO3 entirely to compensation when the PaCO2 exceeds the predicted value. Discriminator: if measured PaCO2 > expected (0.7 × HCO3 + 21) → concurrent respiratory acidosis; if measured PaCO2 < expected → concurrent respiratory alkalosis (classic in a vomiting patient who is also septic or pregnant).

7. Respiratory alkalosis values (pH High, both Low)

WHAT YOU SEE

Respiratory alkalosis: pH High, HCO₃ Low, PaCO₂ Low

WHAT IT ENCODES

Primary fall in PaCO₂ (hyperventilation) raises pH (alkalemia); the kidney compensates by excreting HCO₃, lowering it. Here pH and PaCO₂ move in OPPOSITE directions (PaCO₂ down, pH up), which fingers PaCO₂ as the PRIMARY problem and HCO₃ as the compensation. Causes: anxiety/pain, hypoxemia (PE, pneumonia, high altitude), CNS lesions, pregnancy (progesterone), sepsis, and early salicylate toxicity. Salicylate poisoning classically produces a mixed primary respiratory alkalosis + high-anion-gap metabolic acidosis.

BOARD TRAP

WRONG answer: misreading a respiratory alkalosis with low HCO₃ as a primary metabolic acidosis. Discriminator: in primary metabolic acidosis pH and PaCO₂ move in the SAME direction (both low with pH low); in respiratory alkalosis the pH is HIGH despite the low HCO₃ — the pH direction tells you which variable is primary.

8. Respiratory alkalosis — acute

WHAT YOU SEE

Acute: HCO₃ will ↓ 0.2 mmol/L per mmHg ↓ in PaCO₂

WHAT IT ENCODES

Acute respiratory alkalosis: HCO₃ falls only ~0.2 mmol/L per 1 mmHg fall in PaCO₂ (roughly a 2 mmol/L drop per 10 mmHg fall), reflecting immediate intracellular/tissue buffering before the kidney engages. Example: acute PaCO₂ fall from 40 to 20 → HCO₃ drops ~4 (24 → ~20). This small HCO₃ fall is why acute hyperventilation can produce a strikingly alkalemic pH.

BOARD TRAP

WRONG answer: calling a respiratory alkalosis "acute" when HCO₃ has fallen far more than the 2-per-10 rule predicts. Discriminator: if the HCO₃ drop exceeds the acute prediction it is either CHRONIC respiratory alkalosis (renal compensation, ~4–5 per 10) or a superimposed metabolic acidosis — check the clinical timeline (minutes/hours vs days).

9. Respiratory alkalosis — chronic

WHAT YOU SEE

Chronic: HCO₃ will ↓ 0.4 mmol/L per mmHg ↓ in PaCO₂

WHAT IT ENCODES

Chronic respiratory alkalosis: after 2–3 days the kidney reduces HCO₃ reabsorption and ammoniagenesis, so HCO₃ falls ~0.4–0.5 mmol/L per 1 mmHg fall in PaCO₂ (~4–5 per 10 mmHg) — roughly double the acute response. This is the only acid-base state in which renal compensation can return pH essentially to normal (e.g., long-term high-altitude residence, chronic liver disease, pregnancy).

BOARD TRAP

WRONG answer: diagnosing a superimposed metabolic acidosis in a high-altitude dweller whose HCO₃ is low. Discriminator: a near-normal pH with chronically low PaCO₂ and proportionally low HCO₃ (4–5 per 10) is fully compensated CHRONIC respiratory alkalosis — only call it a second disorder if HCO₃ falls more than the chronic formula predicts or the pH is frankly acidemic.

10. Respiratory acidosis values (pH Low, both High)

WHAT YOU SEE

Respiratory acidosis: pH Low, HCO₃ High, PaCO₂ High

WHAT IT ENCODES

Primary rise in PaCO₂ (hypoventilation/CO₂ retention) lowers pH (acidemia); the kidney compensates by generating and reabsorbing HCO₃, raising it. pH and PaCO₂ move in OPPOSITE directions (PaCO₂ up, pH down), identifying PaCO₂ as PRIMARY and HCO₃ as compensation. Causes: COPD/obstructive disease, neuromuscular weakness (Guillain-Barré, myasthenic crisis), CNS/opioid-benzodiazepine sedation, obesity hypoventilation, chest-wall disease.

BOARD TRAP

WRONG answer: reading high HCO₃ as a primary metabolic alkalosis. Discriminator: in metabolic alkalosis the pH is HIGH; in respiratory acidosis the pH is LOW despite the high HCO₃ — again the pH direction names the primary disorder. A patient with low pH, high PaCO₂, and high HCO₃ has respiratory acidosis with renal compensation, not alkalosis.

11. Respiratory acidosis — acute

WHAT YOU SEE

Acute: HCO₃ will ↑ 0.1 mmol/L per mmHg ↑ in PaCO₂

WHAT IT ENCODES

Acute respiratory acidosis: HCO₃ rises only ~0.1 mmol/L per 1 mmHg rise in PaCO₂ (~1 mmol/L per 10 mmHg), from immediate tissue/cell buffering before renal compensation begins. Example: acute PaCO₂ jump from 40 to 80 → HCO₃ rises only ~4 (24 → ~28) and pH drops sharply. Manage by reversing the cause (naloxone for opioids, BiPAP/ventilation for hypoventilation) — bicarbonate is NOT the treatment.

BOARD TRAP

WRONG answer: labeling an acidosis "acute" when HCO₃ has risen well above the 1-per-10 rule. Discriminator: a large HCO₃ rise in apparent acute respiratory acidosis means it is actually CHRONIC (renal compensation has had days to work) OR there is a superimposed metabolic alkalosis — reconcile with the clinical time course before treating.

12. Respiratory acidosis — chronic

WHAT YOU SEE

Chronic: HCO₃ will ↑ 0.4 mmol/L per mmHg ↑ in PaCO₂

WHAT IT ENCODES

Chronic respiratory acidosis: over 3–5 days the kidney maximizes HCO₃ generation so HCO₃ rises ~0.4 mmol/L per 1 mmHg rise in PaCO₂ (~3.5–4 per 10 mmHg) — about 4× the acute response. Classic in stable COPD: PaCO₂ 60 → expected HCO₃ ≈ 24 + 4×(2) = ~32, with pH only mildly low. Giving these patients high-flow O₂ can blunt hypoxic drive and worsen hypercapnia; titrate O₂ to SpO₂ 88–92%.

BOARD TRAP

WRONG answer: over-oxygenating a chronic CO₂ retainer or attributing a very high HCO₃ purely to chronic compensation. Discriminator: HCO₃ above the chronic prediction (3.5–4 per 10) signals a superimposed metabolic alkalosis (e.g., a COPD patient on loop diuretics or with vomiting) — and remember chronic compensation, like all single disorders, never normalizes the pH.